

JANA2

Object-Centric Image Captioning for Indian Video Surveillance

Building a crime and incident detection system that explains itself

This report documents the design, failures, diagnoses and measured results of a surveillance captioning system built from scratch. Every claim here is backed by a number that was measured on this project, not estimated.

Headline results

Outcome	Measurement	Method
Alert quality transformed	separation 0.03 → 0.86	replaced hand-written phrase pairs with learned linear probes
8 of 9 checks now usable	AP 0.80 – 0.95	supervised probes fitted on VLM-verified labels
Encoder 7x smaller	93M → 13.4M, ~3% AP loss	knowledge distillation with a task-aware loss
Whole system inside budget	109M → 29M	against a 100M ceiling
Labels created from zero	45,891 usable frame labels	weak supervision via Moondream3, no human annotation
Thresholds measured	2 false alarms / 1000 frames	calibrated on 400 real CCTV frames

The single most important lesson

Three separate bugs this project had the same shape: **a relative ranking was treated as evidence**. A softmax always elects a winner, so the model reported a scene, a name and an attribute with confidence even when nothing in its list matched. The fix in every case was an **absolute floor** — a raw similarity below which the system says nothing rather than guessing. Ranking answers *which is most likely*; it never answers *is any of this true*.

1 · The problem and the architecture

The system takes a still frame from an Indian surveillance camera — a housing-society gate, a road, a mall entrance, a factory floor — and produces two things: a caption a person can read, and a list of alerts an operator can act on. It must be able to say *why* it said what it said.

Why there is no language model in the pipeline

The predecessor project used a caption decoder and failed for a specific reason: the decoder overrode correct evidence. The perception stage identified a cat; the decoder wrote "dog", because that sentence was more probable. Fluency was optimised, truth was not.

JANA2 removes the decoder entirely. Every word in the output traces to a rule that can be read in source. The cost is prose that is merely serviceable. The benefit is that a wrong word can always be traced to a specific line — which is the property a security system actually needs.

JANA2 pipeline — every stage inspectable, no language-model decoder

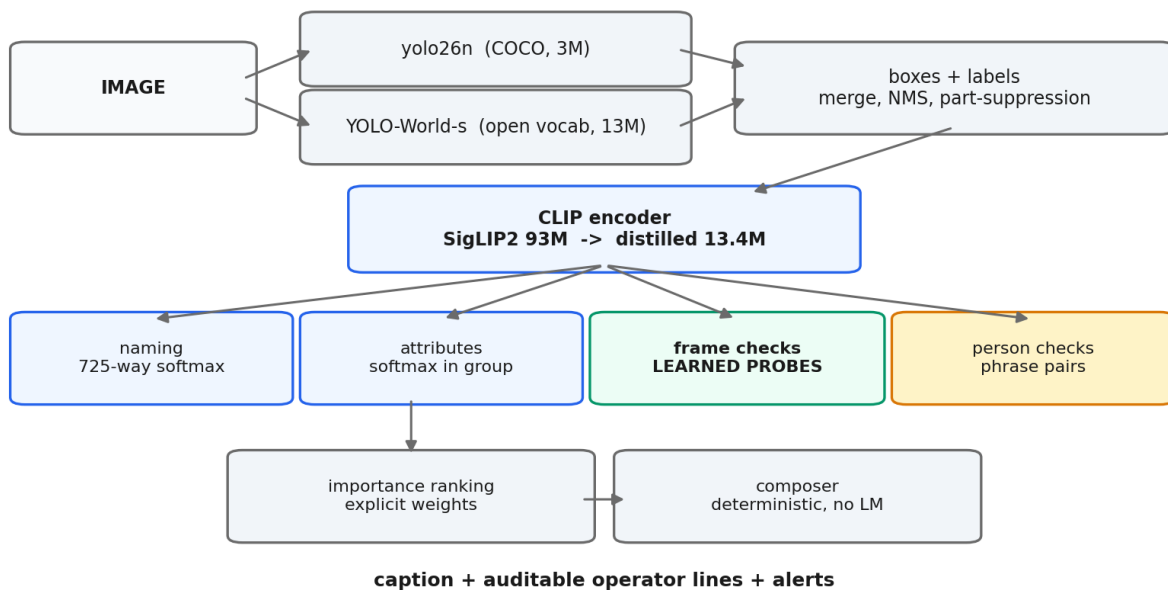


Figure 1 — Detectors supply boxes and coarse labels; CLIP supplies every judgement; a deterministic composer assembles the output.

The three readouts

One early insight shaped everything after: **different questions need different mathematics**, and using the wrong one produces confident nonsense rather than an error.

Question	Readout	Why
What is this object?	softmax over 725 names	many mutually exclusive options compete
What colour is the shirt?	softmax within a group	only ~15 options are relevant
Is there a fire?	contrastive pair, then probe	a yes/no claim against its negation

What went wrong before this was understood

Naming through a sigmoid gave near-zero scores for correct answers. Yes/no checks scored through SigLIP's native logit scale (~100) turned a cosine gap of 0.05 into 99% confidence — a studio portrait produced *no helmet* 99%, *crowd surge* 87% and *darkness* 92% simultaneously.

2 · How the work progressed

Project phases

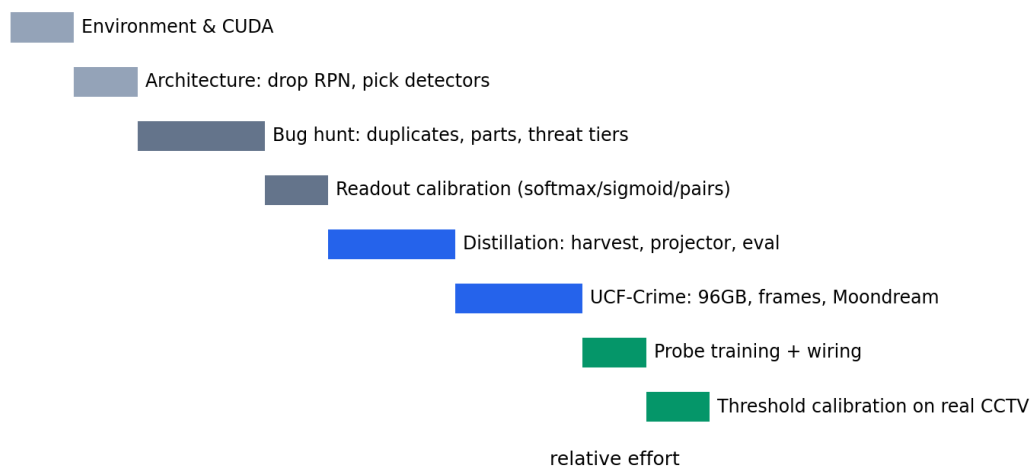


Figure 2 — Roughly half the effort went to diagnosing failures rather than adding features, which is normal and worth expecting.

Phases in detail

Phase	What happened	What it produced
Environment	CUDA 12.4 driver vs newer builds; torch reinstalled repeatedly	Reduced a constraint that shaped every later tool choice
Architecture	Dropped the class-agnostic RPN; settled on yolo26n + FLOPs + Moondream	50% fewer boxes, fewer bad zero-shot guesses
Bug hunt	Duplicate detections, body parts as entities, flat weapon classes	Symmetry-first merging, part/entity split, tiered threat model
Readout fixes	softmax vs sigmoid vs contrastive pairs; gating; temperature	False alarms on portraits eliminated
Distillation	154k crops harvested, projector trained on frozen backbone	Backbone encoder replaced by 13.4M
Weak supervision	96GB UCF-Crime, 56k frames, Moondream3 via vLLM	45,891 frame-level labels from zero human annotation
Probes	Logistic probes fitted per check, gated on AP	8 checks moved from guesswork to measurement
Calibration	400 real CCTV frames, thresholds at a false-alarm budget	Thresholds derived from data instead of intuition

3 · Failures, diagnoses and fixes

This is the most instructive section. Each of these cost real time, and the diagnosis mattered more than the fix.

3.1 · Every check reported 61%

Symptom	Unrelated checks all fired at ~61% on ordinary images.
Root cause	$\text{prob} = \text{sigmoid}(\text{margin} \times T)$ returns 0.5 when there is no evidence at all $$. A threshold of 0.60 therefore fired on a c
Fix	Rescaled so no-evidence reads 0.0, and retuned temperature to match.
Lesson. Know where your scale starts. A probability that is 0.5 for 'nothing here' will fire on nothing.	

3.2 · A parked motorcycle was 'a road accident scene'

Symptom	Scene classification asserted an accident on an empty driveway.
Root cause	Scene is a softmax over 40 labels: it always elects a winner. With no matching place in the list, the nearest one w
Fix	Added a raw-cosine floor; below it the caption says nothing about location. Also removed the event label from a vocabulary
Lesson. A forced choice among wrong options still returns an answer. Absolute evidence, not rank.	

3.3 · A dustbin became a petrol can — and then a weapon

Symptom	Ordinary bins raised incendiary threat alerts.
Root cause	Not a CLIP failure. 'dustbin' was absent from the detector's class list $$ while 'petrol can' was present, so the nearest
Fix	Added benign lookalikes for every weapon class; raised the trust threshold to 0.35.
Lesson. An open-vocabulary class list is a forced choice. Whatever you leave out is assigned to whatever you left in.	

3.4 · Margins of 0.03 — the checks were never working

Symptom	Alerts were either constant or absent; no threshold satisfied both.
Root cause	Four of five negative phrases for [fire] contained the word 'fire'. CLIP has no representation of negation $$ — 'a scen
Fix	Rewrote every negative to describe what IS present. Margins doubled to tripled.
Lesson. Know what your model cannot represent. Negation is a classic blind spot in contrastive models.	

3.5 · Constrained decoding produced clean, wrong answers

Symptom	Forcing a VLM to answer only 'yes' or 'no' returned 'no' to everything.
Root cause	The token bitmask collapsed the distribution rather than steering it — including on questions the model answered correctly
Fix	Free-form generation with a three-way parser; unreadable answers return null and are dropped.
Lesson. Clean output format is not correctness. A constraint that improves parsing can destroy content.	

4 · Knowledge distillation

The encoder was 93M of a 100M budget — 85% of the cost for one component. SigLIP2 carries capacity for artwork, food and celebrities; this system sees streets, gates and vehicles.

Knowledge distillation — teacher embeddings ARE the labels (no human annotation)

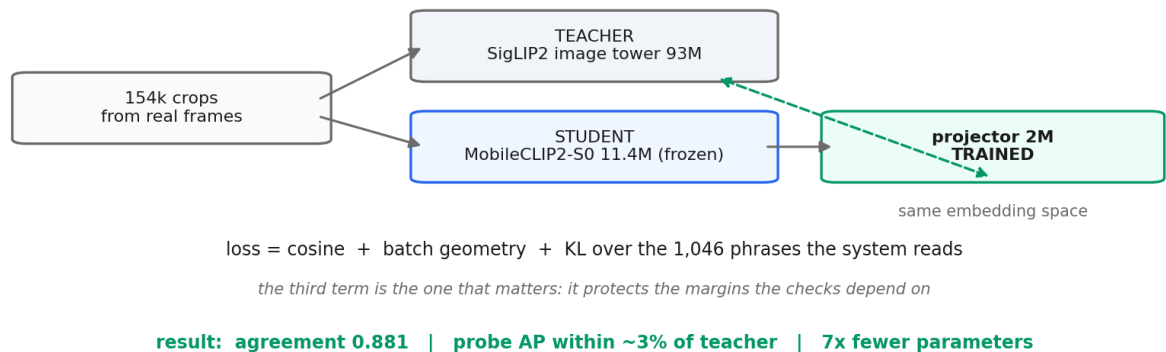


Figure 3 — The teacher's own embeddings are the training targets, so no human labels are involved and every image is usable training data.

Three decisions that made it work

Decision	Reasoning
Inherit, never start from scratch	Random initialisation needs millions of images; a pretrained small tower needs tens of thousands.
Replace only the image tower	The student is trained INTO the teacher's space, so all cached text embeddings, thresholds and vocabulary are preserved.
Optimise what the pipeline reads	Nothing downstream consumes a raw embedding — it consumes margins between near-identical phrases.

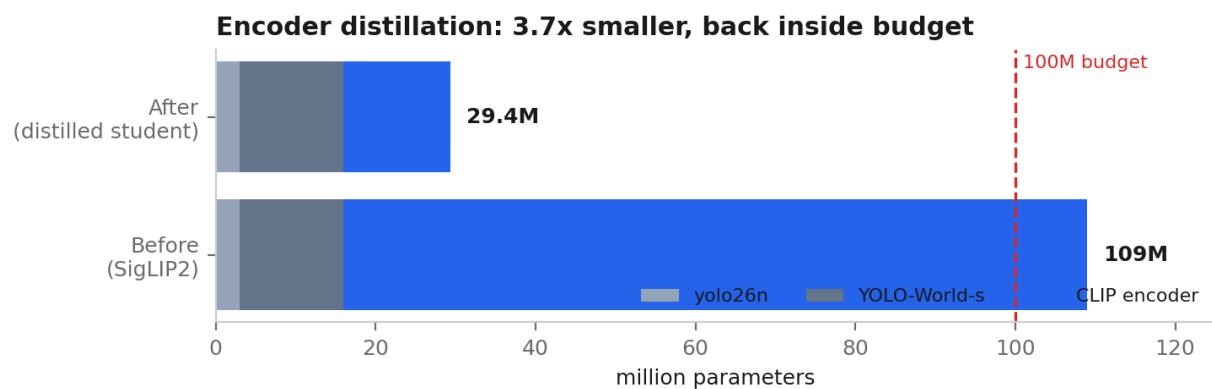


Figure 4 — Detector size was never the problem; the encoder was.

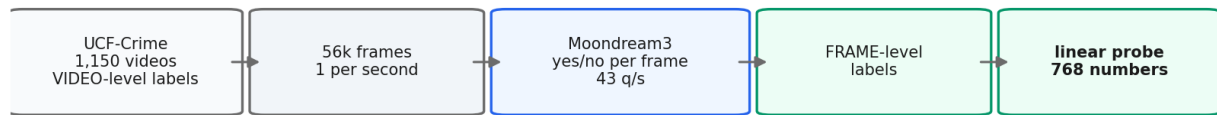
The measurement that mattered

Cosine agreement with the teacher was 0.881, which sounded mediocre. But probe accuracy — what the system actually does — came within ~3%. **Agreement measured similarity; AP measured whether it still works.** A probe is fitted to whatever geometry the encoder produces, so it needs the classes to stay separable, not the embeddings to be identical.

5 · Creating labels where none existed

The project had no labelled evaluation data of any kind. Every threshold was set by intuition and adjusted by looking at single screenshots — which is how the same parameter got moved in both directions in one day.

Weak supervision: a VLM converts video-level labels into frame-level ones



"this video contains a fight" does not say WHICH FRAMES.

Frames from a fight video with no fight visible become HARD NEGATIVES —
same camera, same place, no fight. Nothing is discarded.

Figure 5 — UCF-Crime provides 1,900 real CCTV videos labelled by category. A vision-language model converts those video-level labels into frame-level ones.

Why the non-events are the valuable part

A four-minute Fighting video contains perhaps fifteen seconds of fighting. Frames from that video showing an empty corridor are not waste — they are **hard negatives**: same camera, same lighting, same place, no fight. A probe trained against those learns the event. Trained only against unrelated normal footage, it could learn 'grainy indoor corridor' instead.

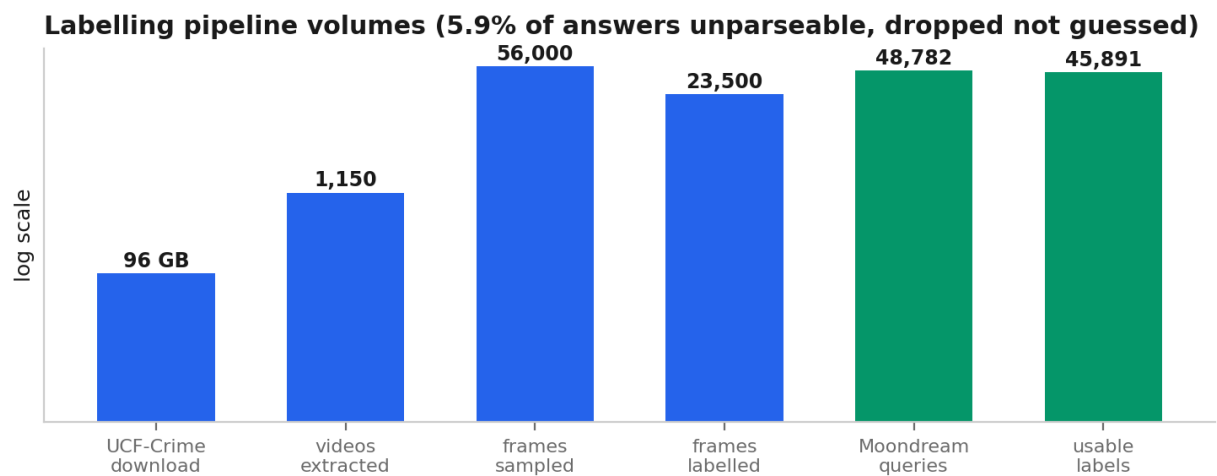


Figure 6 — 5.9% of answers were unparseable. Those were written as null and excluded, never guessed: a wrong label is worse than a missing one.

Engineering detail	Value	Why it mattered
Batched inference	43 q/s vs ~1 q/s	one-at-a-time discards the whole point of vLLM
Warmup pass	—	JIT compilation on first inference returned empty answers
Resume-safe output	—	a 20-minute job that cannot resume is a 20-minute job you rerun
Null on unparseable	5.9% dropped	silent misparsing would have poisoned every probe

6 · Results

From hand-written phrases to fitted vectors

A check was a comparison against sentences someone wrote. A probe is a single 768-number vector fitted to real examples — same dot product, same speed, same inspectability, but optimised for the decision instead of hoping a sentence lands in the right place.

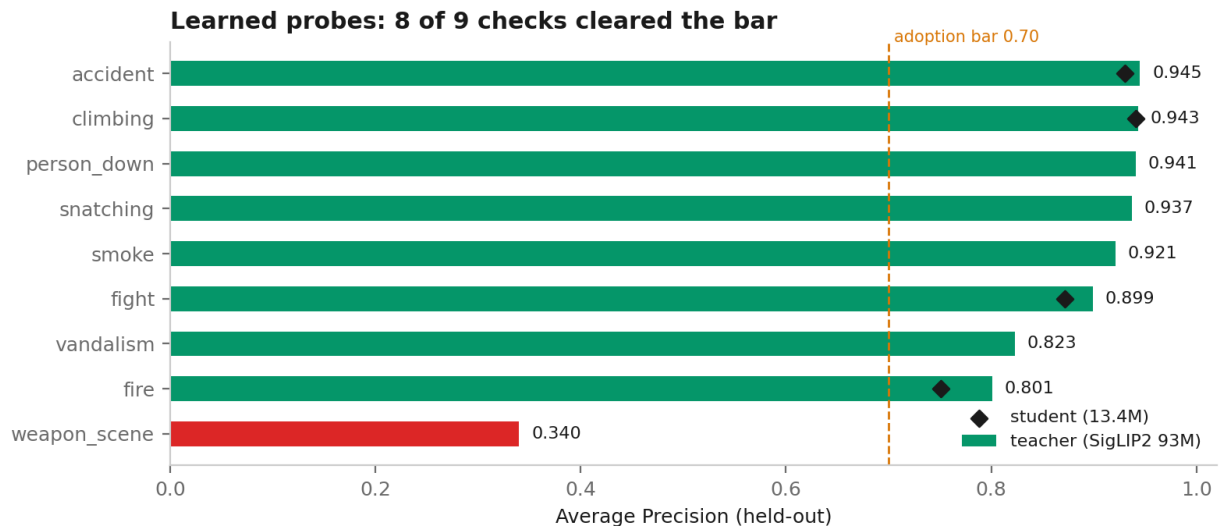


Figure 7 — Average Precision on held-out frames. The baseline is the positive rate, not 50%: with 2.6% positives, always answering 'no' scores 97% accuracy and detects nothing.

	before (phrases)	after (probes)
separation, positives vs negatives	0.02 – 0.05 cosine	0.62 – 0.87 probability
thresholds	guessed, moved twice in one day	measured at a false-alarm budget
checks usable	unclear — no measurement existed	8 of 9, with AP stated

The check that failed, and why that is a good outcome

weapon_scene reached AP 0.34 — only 246 positives out of 9,289, because weapons are small and rarely legible in CCTV. It was rejected automatically by an AP gate and kept its phrase pair. A system that reports which of its components do not work is more useful than one that reports uniform success.

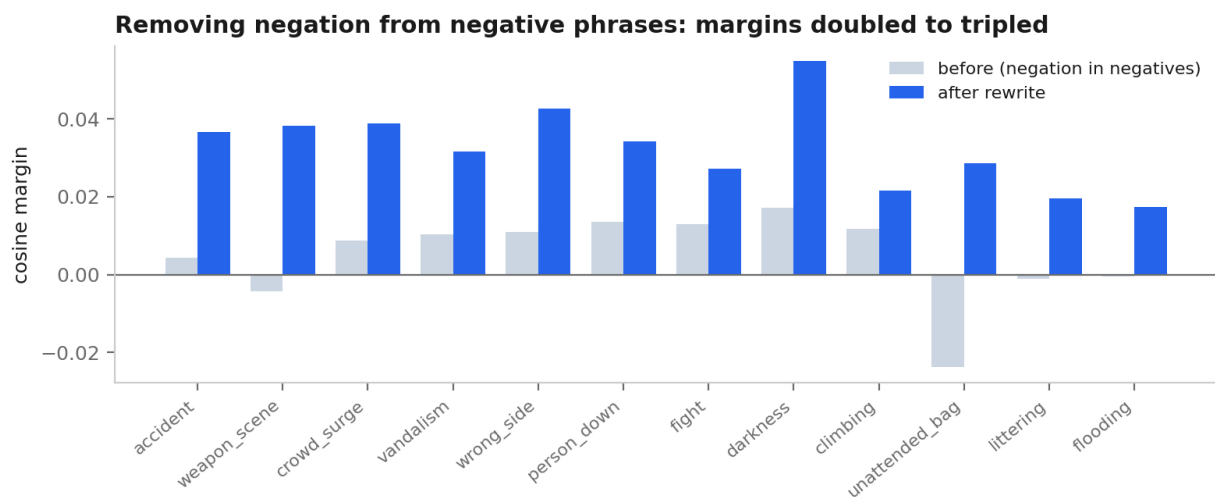


Figure 8 — The effect of removing negation from negative phrases, measured on the same 800 held-out crops before and after.

7 · Concepts used, and what they mean

Written as an internship reference — the ideas worth carrying to the next project.

Vision–language models

Term	Meaning in this project
CLIP / SigLIP	Two encoders trained so an image and its caption land near each other. Lets you classify by describing classes
Zero-shot	Using a model on a task it was never trained for by describing the task. Convenient, and weak on compositional
Embedding space	The shared coordinate system. Two encoders are only comparable inside the same one, which is why every ca
Prompt ensembling	Averaging several wordings of the same concept to cancel the noise in a single text embedding.

Making decisions from similarity

Term	Meaning in this project
Cosine margin	best-positive minus best-negative similarity. This project's lived in a 0.02–0.05 band — the reason nothing work
Temperature	Controls how sharply small differences become confident percentages. SigLIP's native value (~100) turns noise
Absolute floor	A minimum raw similarity below which the system declines to answer. The fix for three separate bugs.
Gating	Refusing to run a check unless its preconditions hold — no helmet check without a motorcycle.

Supervised learning on frozen features

Term	Meaning in this project
Linear probe	One vector fitted on top of a frozen encoder. Cheap, inspectable, and dramatically better than hand-written phra
Class imbalance	With 2.6% positives, 'always no' is 97% accurate. Handled with pos_weight in the loss and measured with Aver
Average Precision	Area under precision–recall. Baseline is the positive rate, not 50%. Cannot be gamed by predicting the majority
Hard negatives	Examples that look like the target but are not. Worth far more per sample than easy ones.
Held-out split	Fitting and judging on the same data measures memorisation. 20% was withheld throughout.

Model compression

Term	Meaning in this project
Knowledge distillation	Training a small model to imitate a large one. The teacher's outputs are the labels, so no annotation is needed.
Weight inheritance	Starting the student from a pretrained small model rather than randomly. Cuts the data requirement by an order
Projector	A small trained map from the student's space into the teacher's. Because the backbone stays frozen, features a
Task-aware loss	Optimising the quantity the system consumes — here, the similarity distribution over the project's own vocabula

Weak supervision

Term	Meaning in this project
Weak / noisy labels	Labels that are correct at a coarse level (this video contains a fight) but not at the level you need (which frames
VLM as annotator	Using a vision-language model offline to convert coarse labels into fine ones. It runs once and is not part of the
Label noise handling	Unparseable answers written as null and excluded, rather than defaulted. A wrong label is worse than a missing

Systems and engineering

Term	Meaning in this project
CUDA compatibility	Driver version caps which builds run. A forward-compatibility library resolved a full day of blocked work — worth

Term	Meaning in this project
Environment isolation	The labelling stack lived in its own venv and communicated by JSONL, so it could not break the working one.
Batched inference	43 q/s vs ~1 q/s for the same model. Throughput is an architectural choice, not a tuning detail.
Caching with identity stamps	Every cached artefact records which encoder produced it and refuses to load against another.
Regression testing	Freezing outputs and diffing after a change — the only way to notice that fixing one thing broke another.

8 · What this project actually taught

Measure before tuning

A full day went into adjusting thresholds by looking at individual screenshots. The same parameter was moved in both directions within hours. The underlying problem was that the signal being thresholded was 0.03 wide and nobody had measured it. **When tuning is not converging, the parameter is usually not the problem.**

Distinguish ranking from evidence

Softmax answers 'which is most likely'. It never answers 'is any of this true'. Three separate bugs — scene, naming, attributes — were the same mistake in different places.

Failure modes matter more than accuracy

Constrained decoding produced perfectly formatted answers that were wrong. A silent parsing bug would have marked every frame negative and looked like a data problem. **Prefer systems that fail loudly.**

Negative results are results

The RPN was dropped, weapon_scene was rejected on measurement, and the class-agnostic proposal path was abandoned. Each of those decisions was backed by numbers, and each saved more time than it cost.

Architecture can compensate for a weaker model

A 13.4M encoder performs nearly as well as a 93M one here, largely because the system trusts detector labels over zero-shot naming, uses grouped softmax for attributes, and gates checks on preconditions. A naive CLIP-only captioner would have collapsed at that size.

What remains

Item	Why it matters	State
Person-crop probes	All 8 person checks are still phrase-based; weapon_held misfires on phrase written	open
Recall measurement	False-alarm rate is measured; catch rate is not	data ready
Attribute floors	Same ranking-as-evidence bug, third location	identified
Regression test set	94 prompts written for generated test images	unused
weapon_scene	Needs more visible-weapon positives than CCTV provides	open

Closing note

The system began the project with alerts that fired on everything or nothing, thresholds nobody could justify, and no way to tell whether a change helped. It ends with eight checks whose accuracy is stated as a number, thresholds derived from real footage, an encoder a seventh the size for a measured 3% cost, and a labelled dataset built from scratch. **The largest single improvement came from replacing sentences a human wrote with vectors fitted to data.**